Kunal Jindal

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EDUCATION

International Institute of Information Technology, Bangalore *Aug 2023 – Jul 2028*

BTech + MTech (Dual Degree) in Computer Science *CGPA: **3.7 / 4.0***

MENTORSHIP

Teaching Assistant, Data Structures and Algorithms Lab *Jan 2026 – May 2026*

- Mentored **150+ students** on core concepts in Data Structures, Algorithms, and Algorithmic Thinking.

ACHIEVEMENTS

- **Codeforces Expert** (**Peak Rating: 1708**): Regular participant in high-difficulty algorithmic contests.
- **IICPC 2026 Codefest Regionalist**: Ranked **278** among **13,000+** participants in preliminary rounds.
- **Turing Cup**: Secured **21st** out of **1,000+** competing teams.
- **LeetCode Knight** (**Rating: 1881**): Placed in the top 5% of LeetCode users.
- **Dean's Merit List**: Awarded for maintaining a **CGPA of 3.6+ over two consecutive academic years**.
- **Grand Finalist, Meta OpenEnv Hackathon**: Qualified for the on-site Grand Finale and ranked in the **Top 100** out of **30,000+** competing teams nationwide.

PROJECTS

NASA Log Analytics – ETL Pipeline | *Java, Hadoop, Pig, Hive, MongoDB* | *Apr 2026 – May 2026*

- Built a unified Java CLI ETL tool processing **3.4M+** NASA HTTP logs via **4** swappable engines (MapReduce, Pig, Hive, MongoDB), using shared regex across 2 batching strategies (record-count, file-based).
- Benchmarked **12** query-pipeline combinations, with MapReduce running **11×** faster than Hive (54s vs. 610s).
- Designed 6-table **MySQL** schema with JDBC batch inserts; implemented **compare** command verifying byte-identical results.

Code-Specific RAG Prompt Compressor | *Python, NLP, RAG* | *Apr 2026 – May 2026*

- Developed a code-specific prompt compressor for RAG bug-fixing using **AST-based** type-aware token ranking (JavaParser), addressing limitations of entropy-based methods (LLMLingua) that ignore syntactic role of code tokens.
- Fine-tuned **CodeT5 (770M)** with pointer-generator on **471K** triples across 9 ratios from Bugs2Fix; deployed to HuggingFace.
- Achieved **87%** token reduction on CodeLlama-34B with zero quality loss (CodeBLEU **88.39** vs. **88.30** baseline).

Doctor Appointment Booking System | *Node.js, Express, MongoDB* | *Jun 2025 – Jul 2025*

- Engineered a healthcare REST API with **JWT** access/refresh token rotation, server-side token blacklisting via MongoDB TTL indexes, and reusable middleware for role-based authorization and input validation across 3 user roles.
- Designed 4 MongoDB aggregation pipelines for admin analytics and scheduled cron jobs for automatic appointment expiry.
- Validated **17** API workflows via integration tests covering auth, booking conflicts, and cascading cancellations.

Multi-Client Course Registration Server (CLI) | *C, UNIX System Calls* | *Apr 2025 – May 2025*

- Architected a TCP client-server system in C using UNIX socket APIs and **fork()** to handle **10** parallel client connections.
- Developed a custom flat-file database engine with 6 tables and **file locking** (read/write/unlock) for concurrent data safety.
- Implemented CRUD operations over sockets for 3 user roles (student, faculty, admin) with role-based access control.

TECHNICAL SKILLS

Languages: C, C++, Python, Java, SQL

Systems & Backend: REST APIs, Client–Server Systems, Concurrency, UNIX System Calls

Databases & Tools: MySQL, MongoDB, Hadoop, Apache Pig, Apache Hive, GitHub

RELEVANT COURSEWORK

Data Structures and Algorithms, Design and Analysis of Algorithms, Advanced Algorithms, Operating Systems, Database Management Systems, Computer Networks, NoSQL Systems, Machine Learning, Natural Language Processing

EXTRACURRICULAR ACTIVITIES

Member, CAMP (Club for Algorithms, Mathematics and Programming), IIIT Bangalore

- Led ICPC-style contests and discussion sessions on competitive programming for club members.